

Exploring Subjective and Objective Measures of System Effectiveness

Shih-Chu Chen

Institute of Service Science, National
Tsing Hua University, Taiwan
shihchu.chen@iss.nthu.edu.tw

Ko-Jen Chiu

Institute of Service Science, National
Tsing Hua University, Taiwan
karen.chiu@iss.nthu.edu.tw

Pei-Fang Hsu

Institute of Service Science, National
Tsing Hua University, Taiwan
pfhsu@iss.nthu.edu.tw

ABSTRACT

The effectiveness of information systems (IS) is essential for evaluating user performance between their behavior and computer. However, the effectiveness evaluation poses challenges due to the multidimensional nature of measures, especially in evaluating business intelligence systems (BI). This study aims to investigate measures from both subjective and objective of BI effectiveness and classify BI effectiveness into five dimensions based on the IS maturity model, including *Operate*, *Consolidate*, *Integrate*, *Optimize*, and *Innovate*. The study compares results obtained through self-reported with computer-recorded measures to gain a comprehensive understanding of system utilization. This research received 1480 questionnaires, accompanied by the collection of 1-year log data reflecting the actual system usage of each respondent. The findings reveal a partial alignment between self-reported and actual usage frequency depending on divergent effectiveness levels. These insights contribute to the complex dynamics underlying user behavior and provide practical implications for organizations seeking to enhance effective BI utilization.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); HCI design and evaluation methods; • **Information systems** → Information systems applications.

KEYWORDS

Effectiveness, Subjective and Objective Measures, User Behavior, Business Intelligence

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1 INTRODUCTION

The effectiveness evaluation of the information system (IS) plays a critical role in the association between users and systems, especially in the estimation of presenting users' behavior and their

performance. As highlighted by DeLone and McLean (2016), evaluating the effectiveness of IS poses a significant challenge for both researchers and practitioners [10], particularly the well-known system for supporting decision-making in firms – business intelligence (BI) system [16, 23]. Despite practitioners acknowledging the importance of measuring the effectiveness of their IS investments, the required diverse analysis skills of this high-level BI system increase the difficulty of the evaluation [11]. Different observers may consider different variables when assessing the effectiveness and may have diverse criteria for evaluating each variable.

Among various measures of effectiveness, system usage has been one of the most frequently proposed measures [1, 9, 24, 27]. The results of Straub et al. (1995) suggest that “system usage should be factored into self-reported system usage and computer-recorded system usage,” knowing as subjective and objective measurements [22]. However, methods for measuring IS effectiveness are often one-dimensional and over-simplified [18, 27]. A prevailing scenario unfolds where managers and executives primarily gauge the effectiveness of a system by solely focusing on system usage metrics, particularly the volume of log data generated. Some studies merely relying on computer-recorded data overlook other vital aspects of system effectiveness, such as the level of user involvement [5], perceived impact such as task productivity and task innovation [25], and employees' IT competencies [26]. In contrast, other studies depend on self-reported data and highlight the advantages and accuracy of presenting users' perspective and intention [30]. Consequently, this limited view can lead to misguided assessments and decisions, as the true effectiveness of BI systems is not fully understood.

To capture a comprehensive picture, a holistic approach incorporating both subjective and objective measures is crucial. This research aims to explore subjective and objective perspectives on BI system, investigating factors influencing system effectiveness. The paper addresses the research question: How is the relationship between self-reported (subjective) and computer-recorded (objective) measures for BI effectiveness? By answering this question, we contribute to providing a more nuanced and comprehensive evaluation of multi-dimensional BI system effectiveness from actual firm usage behavior.

2 LITERATURE REVIEW

2.1 Subjective and Objective Measures

In the exploration of information system (IS) effectiveness, the assessment of usage often incorporates both subjective and objective measures. Objective measures rely on tangible and observable data, typically collected through devices like system logs or sensors that

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record physical activities; while subjective measures involve capturing individuals' perceptions, opinions, experiences, and memories related to their physical activity levels.

Some studies such as Aranda et al. (2014) found that objective measures tend to be more reliable than subjective measures, suggesting that subjective measures may be prone to bias [3]. In contrast, some studies show that subjective measures have more advantages than objective measures in some contexts. In the study of Zhao (2011), suggesting that subjective measures of car dependence are important as they reveal subjective measures enhance the understanding of people's intent to reduce car use [30]. Muckler and Seven (1992) also highlighted the subjectivity of a human as part of scientific and technological measurement [15]. They argued that subjective measures are valuable because they capture individual differences in abilities and attitudes that may not be captured by objective measures of performance. The diverse findings from prior studies underscore the complexity of the relationship between subjective and objective measurements.

2.2 Perspectives on the Relationship Between Subjective and Objective Measures

In exploring the strengths and weaknesses between subjective and objective measures, several studies have aimed to compare these two types, seeking insights into their similarities and differences. However, the literature presents varying perspectives on the alignment between subjective and objective measures.

Some studies reported a notable alignment between subjective and objective measures [2, 17]. Straub et al. (1995) and Barnett et al. (2006) both found that self-reported and computer-recorded usage is consistent but lacks a robust relationship [4, 22]. Conversely, other research reveals discrepancies between individuals' subjective interpretations and evaluations of objective measurements. One discovered no perfect match between self-reported and actual usage in a mobile application system [29]. Furthermore, the case study demonstrated significant discordance between subjective and objective assessments, emphasizing that users' opinions are influenced by factors beyond its primary objective performance measures [13]. These factors may include limitations in personal recall [20] or the social impact within departments [6, 28].

Despite research comparing self-reported and computer-recorded results in fields like human-computer interaction (HCI), information systems (IS), and medical devices, few studies specifically investigate the topic of effectiveness measurements. Therefore, the overarching goal is to discern how both measures can be effectively utilized in assessing the effectiveness of BI systems, and our study endeavors to evaluate this relationship comprehensively.

2.3 Business Intelligence Maturity and Effectiveness Evaluation

The evaluation of system effectiveness is complicated and challenging due to its multidimensionality, including quantitative and qualitative aspects [12], behavioral and perceptual variables [24], and the multiple and often conflicting evaluator viewpoints [21]. To be more comprehensive in evaluating BI effectiveness, the BI maturity model was proposed to categorize thorough levels of usage behavior. Evolved from the Information Evolution Model (IEM)

[8], organizations can be classified into five different levels of how maturely they use BI from low to high levels (operate, consolidate, integrate, optimize, and innovate) [7], as shown in Table 1. Building from the BI maturity, this study considers these five levels to measure BI system effectiveness.

In summary, the results of the literature review show only a limited number of studies have undertaken comprehensive measurement and consideration of the multiple effectiveness dimensions. It also emphasizes the current challenges the field is facing in understanding the effective use of BI and determining suitable approaches to measure system use.

3 METHODS

3.1 Study Design and Measurements

This study collected both subjective and objective data to understand BI effectiveness. A questionnaire was designed to investigate employees' self-reported BI usage (see Appendix A.1). The questionnaire consisted of two sections. The first section aimed to gather basic information about each respondent, including demographics such as gender, age, sector, and their experience with BI-related resources. In the second section, respondents used the seven-point Likert scale to express the degree of frequency (ranging from 1 = never to 7 = very often) for their BI system usage across five distinct contexts. Each context corresponded to different levels of BI effectiveness, aligning with the IS Maturity Model [7, 19]. These questions guided respondents to answer their behavior at the *Operate* level as generating meeting reports, at the *Consolidate* level as analyzing departmental data, and at the *Integrate* level as integrating data across functional units. The contexts also encompassed more advanced usage scenarios, such as creating interactive dashboards at the *Optimize* level and integrating BI with robotic process automation (RPA) at an *Innovative* level. An optional open-ended question was presented at the end of the questionnaire to collect employees' qualitative feedback regarding their experience in using BI.

3.2 Data Collection and Analysis

In this research, data was collected from a corporation that implemented Power BI as an analytical tool for its employees to utilize in their work. The survey was conducted in February 2023, with employees receiving survey participation invitations through the company's email system. Employees who have used Power BI systems are qualified respondents and were eligible to take the survey voluntarily. A total of 1480 responses were collected, representing individuals from 123 functional units categorized in 8 main departments within the organization. From the descriptive statistics, the significant proportion (80%) reported no prior experience in developing BI reports. Additionally, 63 of the respondents voluntarily shared their ideas about BI usage, and we used their responses to complement our quantitative results in the Discussion section. Apart from the self-reported survey data, this study also collected the one-year (March 2022 to February 2023) log data from the Power BI system for all the 1480 employees who participated in our survey. This dataset comprises a total of one million records and access through 1730 reports. For each record, the data represents a user's behavior of accessing a BI report page. In this manner, the results

Table 1: Five-level business intelligence maturity model

Level	Scope	Definition
1. Operate	Individual	Tackle day-to-day issues, generate personal reports or personalized spreadsheets.
2. Consolidate	Department	Integrates and stores information at the departmental level, but not across departments.
3. Integrate	Enterprise	Manage data between employees from various departments.
4. Optimize	Strategy	Employees with intellectual skills, typically uses software programs with automated data analysis techniques to detect relationships in the data and provide predictive results.
5. Innovate	Sustainable growth	Employees require expertise in advanced decision-making, generate an automated exception reporting when something unusual occurs.

Table 2: Descriptive statistics of total and monthly system access records

	N	Total System Access			Monthly System Access		
		Mean	Median	SD	Mean	Median	SD
All Respondents (ALL)	1471	389.7	132.0	778.6	35.8	13.8	68.6
Departments							
Administration (AD)	304	320.0	99.0	532.2	28.5	10.3	44.7
Factory1 (F1)	319	297.7	135.0	532.3	27.6	13.3	45.6
Factory2 (F2)	343	515.3	181.0	954.2	47.9	18.5	89.8
Factory3 (F3)	310	399.9	137.5	773.8	36.4	13.6	65.3
Factory4 (F4)	111	230.2	96.0	347.9	24.9	14.3	31.5
Human Resources (HR)	31	121.7	31.0	228.0	11.0	2.8	19.3
Production (PD)	18	2605.4	1515.0	2399.6	226.6	150.6	195.7
Technology (IT)	35	115.4	77.0	116.5	10.4	8.0	9.9

of the objective system-recorded log data and subjective survey-based data can be compared. To explore the complex relationship between self-reported and computer-recorded usage, visualization and correlation analysis was performed.

4 RESULTS

4.1 Objective Measure: Computer-recorded Frequency

To address potential outliers in the dataset and to ensure more reliable results, a filtering process was applied to exclude nine employees with total system access exceeding 100,000 records. For the requirement of working or monitoring, these extreme records represented auto accessed through computer settings, such as the scheduled reloading report page. This filtering resulted in 1471 valid respondents with paired survey- and one-year BI system access data.

In the analysis of the computer-recorded data, the mean of total access for employees over one year was 389.7 times (*Median* = 132.0, *SD* = 778.6). To obtain objective frequency, the total access was divided by the number of working months, resulting in an average of 35.6 times monthly access (*Median* = 13.8, *SD* = 68.6). Additionally, we summarized both total and monthly access among eight departments. The above-mentioned are organized in Table 2.

Upon examining departmental differentials, observable variations in the mean frequency of system access emerge. Notably, employees within the PD department exhibited a pronounced propensity towards BI usage ($\bar{x}$ = 226.6); conversely, HR and IT employees

displayed comparatively lower reliance on the BI system. This contrast underscored the diverse patterns of BI usage across departments, potentially influenced by job requirements, work purposes, or individual work behaviors. Consequently, evaluating user effectiveness in BI usage becomes inherently challenging and hazed by these departmental differences. Besides, following the prevailing scenario, if managers review solely numerical metrics for behavior, they may introduce bias into low usage. In response to this challenge, we propose addressing the issue through the examination of subjective perceptions from five-level BI effectiveness simultaneously to garner additional insights.

4.2 Subjective Measure: Self-reported Frequency

To understand subjective behavior, employees were asked to report the frequency of their BI usage across five contexts, a scale ranging from 1 to 7. In Figure 1, we found that a score of 4 on the 7-point Likert scale was the dominant answer to most of the usage contexts except for connecting with RPA (S5), in which most of the participants have limited experience.

4.3 Comparison between Subjective and Objective Measures

4.3.1 Correlation Analysis. To further examine the correlation between subjective and objective measures, we categorized the survey respondents into seven groups based on their self-reported scores of 1 to 7 and visualized them using Tableau software. Figure 2 and Appendix A.2 shows the box plots of computer-recorded usage of each group, where the x-axis represents self-reported score

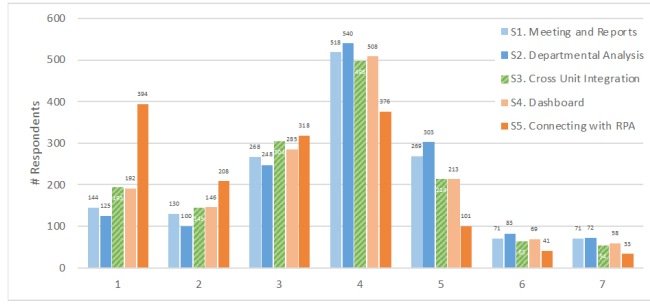


Figure 1: Bar chart of self-reported frequency across each score

from 1 to 7 (subjective measure) and the y-axis shows the number of monthly access (objective measure) corresponding to each employee. From the box plots, there were shown that the median of computer-recorded usage was associated with employees' self-reported usage. In Table 3, in addition, the rows denote varying degrees of BI effectiveness, and the columns correspond to the seven groups (presented as G1 to G7) based on self-reported frequency. Overall, employees with higher self-reported usage had a higher median and mean value of actual system access.

However, some inconsistencies occurred in the difficult BI effectiveness levels (*Optimize* and *Innovate*). To statistically assess the differences between neighboring self-reported scores, independent t-tests were conducted showed in Table 3. For the *Optimize* level, the group with self-reported level G2 ($\bar{x} = 18.1$), which has the lowest computer-recorded access numbers (compared with G3, $t = 1.80^*$). As for the *Innovate* level, the result of connecting with RPA shows insignificant differences among all paired groups, which is inconsistent with the pattern of self-reported. This result suggested that when organizations aim to enhance the advanced levels of BI effectiveness, they should be cautious when relying solely on computer-recorded access data as a measure.

Table 4 shows the correlations between all the subjective measures and objective measures (monthly system access). We noted that the correlation between self-reported and computer-recorded measures for all the levels of effectiveness (S1~S5) is inconsistency.

For example, the correlation between the subjective reported frequency and objective frequency in terms of using BI for departmental data analysis (S2) was 0.370, the highest correlated context with objective measure among five BI effectiveness levels; while the correlation of *Innovate* level (S5) was only 0.076. The results indicated potential bias in measuring effectiveness if firms only examine objective data collected from back-end system.

4.3.2 Correlation Analysis by Departments. In the departmental analysis, we observed diverse correlations between subjective and objective frequency across five levels of BI effectiveness. Notably, the PD department exhibited the lowest correlation between actual behavior and self-reported frequency across all BI effectiveness levels. This result underscored significant diversity in user behavior within the PD department, as denoted by the standard deviation values in Table 2 ($SD = 195.7$). Conversely, AD and HR showcased the highest correlations ($r = 0.215$ to 0.527), indicating a higher degree of consistency between employees' self-reported usage and actual work, especially across the first four effectiveness levels (from *Operate* to *Optimize* Level). Furthermore, F1 to F4, all production-oriented factories, displayed relatively similar correlation values among the first four effectiveness levels (range from 0.178 to 0.358). These findings suggested the work types among departments have differences. For instance, AD mainly focuses on routine usages like meetings, departmental analysis, and dashboard requirements, while HR requires connecting RPA usage for innovative purposes. Thus, the variations in job requirements across different effectiveness levels may influence the connection between self-reported and actual behavior.

4.3.3 Correlation Analysis by Monthly Retraced. To ascertain the timing when the highest correlation occurs with self-reported behavior, a traced-back analysis was conducted. Table 5 displays the correlation between the five contexts (subjective self-reported data) and objective log data. Each column represents an aggregation of the latest monthly access behavior. For instance, the "1m" column captures the system access behavior for the most recent month (February 2023), while the "2m" column reflects the average monthly access from January to February 2023, and so forth. The month-by-month aggregations were calculated to determine the correlation, allowing us to identify the timing that exhibits the highest correlation with self-reported behavior. This analysis provides

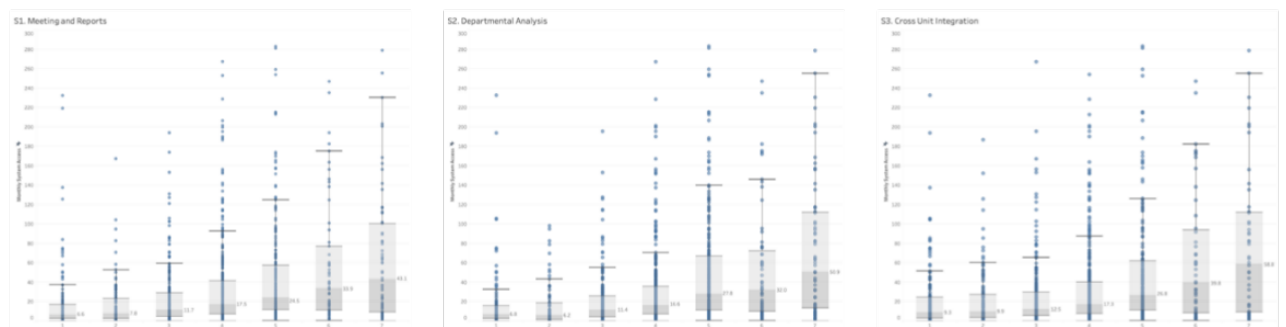


Figure 2: Three box plots examples of monthly system access across each group

Table 3: Monthly system access frequency across each group

BI Effectiveness	Computer-recorded Frequency: Mean (SD)							Significant t-tests Results
	G1	G2	G3	G4	G5	G6	G7	
1. Operate	19.2 (43.5)	19.1 (44.1)	25.9 (57.9)	32.3 (62.2)	51.1 (81.7)	69.9 (108.9)	71.3 (87.6)	$G4\&G5, t = 3.31^{***}$
2. Consolidate	18.3 (43.9)	16.2 (38.1)	21.7 (44.4)	29.9 (64.5)	53.3 (79.2)	66.1 (106.3)	78.7 (87.2)	$G3\&G4, t = 2.07^*$; $G4\&G5, t = 4.38^{***}$
3. Integrate	24.1 (50.5)	19.5 (29.4)	22.1 (31.7)	34.1 (70.7)	58.5 (93.5)	75.8 (113.8)	78.9 (88.4)	$G3\&G4, t = 3.29^{***}$; $G4\&G5, t = 3.42^{***}$
4. Optimize	21.4 (42.9)	18.1 (33.3)	27.0 (68.9)	29.7 (50.7)	58.4 (93.5)	85.5 (116.0)	83.8 (88.0)	$G2\&G3, t = 1.80^*$; $G4\&G5, t = 4.24^{***}$; $G5\&G6, t = 1.76^*$
5. Innovate	36.7 (77.9)	30.7 (48.9)	36.8 (75.8)	31.4 (58.0)	40.6 (67.2)	63.8 (92.9)	49.8 (56.9)	No significant

Note: $^* p < .05$, $^{**} p < .01$, $^{***} p < .001$.

Table 4: Correlation between subjective and objective measures

BI Effectiveness	BI Contexts	Correlation								
		ALL	AD	F1	F2	F3	F4	HR	IT	PD
1. Operate	S1. Meeting and Reports	0.336	0.527	0.289	0.229	0.228	0.319	0.429	0.189	-0.023
2. Consolidate	S2. Departmental Analysis	0.370	0.510	0.358	0.226	0.317	0.292	0.498	0.180	-0.043
3. Integrate	S3. Cross Unit Integration	0.289	0.429	0.302	0.178	0.181	0.232	0.456	0.123	0.050
4. Optimize	S4. Dashboard	0.325	0.488	0.266	0.200	0.273	0.312	0.409	0.038	0.055
5. Innovate	S5. Connecting with RPA	0.076	0.170	0.140	-0.096	0.021	-0.017	0.215	0.014	-0.325

Table 5: Correlation between subjective and objective measures by month-by-month aggregations

BI Effectiveness	BI Contexts	Correlation											
		1m	2m	3m	4m	5m	6m	7m	8m	9m	10m	11m	12m
1. Operate	S1. Meeting and Reports	0.320	0.343	0.361	0.364	0.366	0.359	0.351	0.347	0.341	0.339	0.337	0.334
2. Consolidate	S2. Departmental Analysis	0.348	0.364	0.385	0.391	0.395	0.392	0.386	0.382	0.375	0.375	0.373	0.370
3. Integrate	S3. Cross Unit Integration	0.266	0.276	0.292	0.299	0.302	0.298	0.298	0.297	0.293	0.294	0.292	0.290
4. Optimize	S4. Dashboard	0.273	0.290	0.307	0.315	0.323	0.323	0.326	0.328	0.325	0.327	0.326	0.324
5. Innovate	S5. Connecting with RPA	0.079	0.086	0.092	0.088	0.087	0.080	0.073	0.071	0.064	0.064	0.062	0.060

valuable insights into the dynamic relationship between subjective and objective measures over time.

Various aggregation periods of relative behavior impact the correlation with self-reported behavior differently. For the first three levels (*Operate*, *Consolidate*, and *Integrate*), the highest correlation values were observed with five months of actual behavior. Additionally, at the *Optimize* level, the most correlated behavior was identified when considering eight months of data. For the case of the *Innovate* level, the largest correlation value was considered three months of data. To summarize, the low levels of BI maturity (*Optimize* to *Integrate*) show greatest connection with the last five months of actual behavior, but for the high level of maturity (*Optimize*), it may take eight months to see the consistency.

5 DISCUSSION

5.1 Divergent Consistency among Subjective and Objective Measures in Different Effectiveness Levels

The study delves into comprehending the correlation between employees' self-reported usage and their actual system access in BI across five levels of BI effectiveness. Categorizing employees into groups based on their self-reported frequency reveals a fair correlation, indicating that individuals reporting higher BI usage tend to exhibit increased actual system usage.

However, an exception emerges at the *Innovation* level, where the correlation between subjective and objective frequency measures becomes less evident. Our study explores two potential reasons from literature and from our open-ended questions. First, the introduction of the Robotic Process Automation (RPA) function, characterized by automated system access, can result in elevated

system log activity without direct user intent [14]. This creates a disparity between users' self-reported frequency and the observed actual system access in this specific scenario. Another reason is resonated with the challenges noted by Gudfinnsson and Strand (2017), highlighting that companies often underutilize BI tools for predictive analysis and decision support [11]. Respondents (*R1343*, *R1375*) also reflected the problem of limited knowledge of RPA, stating “*I hope to have RPA courses to learn how to make synergy between RPA and Power BI.*”

Excluding the scenario of connecting BI with RPA, we found that the correlation between self-reported and computer-recorded frequency ranges from 0.289 to 0.370. These correlations slightly surpass the 0.26 correlation reported by Straub et al. (1995) [22]. The findings underscore the significance of employees' self-reported frequency in BI and its impact on their actual system utilization. However, they also emphasize the importance of considering employees' perceived proficiency and engagement with different BI functions when assessing their actual usage patterns.

5.2 Potential Impact among Departments and Time

Our study delves into the disparity between subjective and objective measures, seeking to unravel factors of impacting BI effectiveness. Building on insights from prior studies [6, 28], our exploration centers on departmental differences, aiming to identify notable distinctions. Our results revealed similarities among departments with similar purposes, underscoring the importance of job requirements and departmental characteristics. Some respondents (*R1386*, *R1389*, *R1464*, *R1474*) thought BI have little help on their work, which stated, “*Power BI fundamentally has very limited help for engineers' work.*”

Furthermore, in understanding the temporal aspect of users' perceived usage, we calculate the correlation between self-reported and actual behavior through monthly retracing. Our results unveil that users' perceived system use correlates more strongly within five to eight months of actual behavior. This temporal insight assists managers in determining the appropriate duration for evaluating BI effectiveness as well as for researchers considering suitable data collection period.

In conclusion, these findings caution against relying solely on computer-recorded usage as a comprehensive reflection of BI system utilization. BI effectiveness is multidimensional in nature, emphasizing the complementary nature of subjective and objective measures. While objective measures, such as frequency, may align with subjective assessments for certain aspects, subjective measures provide valuable insights into individuals' perceptions, attitudes, and experiences.

6 LIMITATIONS AND FUTURE WORKS

It is crucial to note that the variability in results can be attributed to a range of factors, including the approach to obtaining perceptual data and measuring usage behavior, the context in which the measurements are taken, and the characteristics of the individuals involved. However, the current objective measurement relies solely on the frequency of access when accessing the system. Future research could explore alternative objective metrics, such as usage

duration, to provide a more accurate reflection of user engagement. Furthermore, we suggest that researchers can concentrate on other cooperation or systems to generalize the findings from investigating objective metrics and standardized measurement. Addressing these limitations and exploring areas for future research will contribute to a more comprehensive understanding of BI effectiveness in various industries.

7 CONCLUSION

This research underscores the significance of embracing both subjective measures, such as employees' self-reported frequency in diverse contexts, and objective measures, like computer-recorded usage, to assess BI system effectiveness. Recognizing that subjective and objective measures may portray an inconsistent narrative, enterprises must refrain from concluding that one measure is definitively right or wrong when discrepancies arise. Instead, different measures capture distinct aspects of effectiveness and can offer valuable insights beyond mere agreement. The holistic approach provides a more comprehensive understanding of the extent of BI effectiveness. This broader understanding allows organizations to shift their perspective and prioritize the strategic utilization of BI systems.

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APPENDICES

A QUESTIONNAIRE

Variables	Values
Demographic	IT Background (IT/non-IT) Power BI Seed User (Yes/No) Have Watched Power BI Tutorial (Yes/No) Have Received Power BI Training (Yes/No) Have Experience in Developing Power BI reports (Yes/No)
Self-reported Frequency (7-point Likert scale)	S1. I apply Power BI in meeting reports (including written, interactive, and oral reports). S2. I apply Power BI in departmental data analysis. S3. I apply Power BI in cross-departmental data integration and analysis. S4. I apply Power BI in creating dashboards (including metrics, key performance indicators (KPIs), and alert systems). S5. I combine Power BI with RPA for automation applications (including automatic scheduling, email reminders, and integration with other systems).
Open-ended	Suggestions or feedback on using Power BI (Optional)

B BOX PLOTS OF MONTHLY SYSTEM ACCESS ACROSS EACH GROUP

